

Document Image Classification Using Lightweight CNNs

Rania Attia, Nourhan Salem, Mariam Youssef

Department of Computer Science, Queen's University • Supervisor: Hazem Abbas • Submitted: May 2026

Abstract—Document image classification is a core task in digital document management, enabling automated categorization of scanned documents such as letters, forms, advertisements, and scientific reports. This paper investigates the use of lightweight Convolutional Neural Networks (CNNs) for document image classification on a 10-class subset of the RVL-CDIP benchmark, with the goal of achieving competitive accuracy under strict parameter budgets suitable for resource-constrained deployment. We compare three approaches evaluated on identical data: a whitespace histogram baseline (25.0%), an OCR+TF-IDF+MLP text pipeline (73.8%), and MobileNetV2 fine-tuned from ImageNet weights (81.07%, $F1 = 0.811$, approximately 3.4M parameters). Results confirm that visual features captured by CNN layers substantially outperform text-based methods on this dataset, particularly for layout-driven categories such as Form and Resume. Per-class analysis identifies Report ($F1 = 0.65$) and Letter ($F1 = 0.79$) as the weakest classes, motivating class-weighted loss and targeted data augmentation in future work. Robustness evaluation under geometric and photometric perturbations, CNN+OCR fusion, and expansion to 14 classes are planned for the final phase.

Index Terms—Document image classification, MobileNetV2, lightweight CNN, OCR, TF-IDF, RVL-CDIP, transfer learning.

I. INTRODUCTION

The digitization of document workflows has produced enormous volumes of scanned images requiring automated organization. Document image classification—assigning category labels based on visual content—underpins digital archiving, automated mail sorting, financial processing, and legal records management. Traditional pipelines rely on Optical Character Recognition (OCR) to extract text before applying text classifiers. This approach struggles with degraded scans, handwritten annotations, and documents where visual layout is more informative than textual content. Convolutional Neural Networks (CNNs) have demonstrated strong performance on image benchmarks, but large architectures (ResNet, VGG, EfficientNet) contain tens to hundreds of millions of parameters, making them impractical for resource-constrained deployment environments such as edge devices, mobile applications, and embedded systems in mail-sorting machines.

This paper investigates lightweight CNNs—specifically MobileNetV2—for document image classification on a 10-class subset of the RVL-CDIP dataset. We establish baselines using whitespace histograms and an OCR+TF-IDF pipeline, both evaluated on the same dataset to enable fair comparison. Our midterm results show MobileNetV2 achieves 81.07% test accuracy with only approximately 3.4 million parameters, substantially outperforming OCR-based methods (73.8%) on identical data. This represents a parameter reduction of over $17\times$ compared to the AlexNet model used in the original RVL-CDIP study [1], while still achieving competitive accuracy on the same benchmark. The lightweight nature of MobileNetV2 makes it suitable for deployment scenarios where computational resources are limited, such as mobile document scanning applications, edge devices in postal sorting facilities, and embedded systems in financial processing pipelines.

A. Project Goals

The goals of this project are fourfold: (1) evaluate a lightweight CNN backbone (MobileNetV2) for document image classification and compare its accuracy and parameter efficiency against large-scale architectures; (2) establish fair OCR-based and feature-based baselines on the same RVL-CDIP subset to enable direct, apples-to-apples comparison; (3) conduct per-class error analysis to identify weak categories and propose targeted improvements; and (4) investigate CNN+OCR fusion strategies and robustness to image perturbations as pathways to further improvement. These goals collectively address the gap between high-accuracy but computationally expensive multimodal models and the practical need for lightweight, deployable document classifiers.

B. Project Scope

The scope of this project encompasses the full pipeline from data preprocessing through model evaluation. We focus on a 10-class subset of RVL-CDIP for analytical tractability, with planned expansion to 14 classes in the final phase. We do not attempt to match the state-of-the-art accuracy of multimodal transformers (e.g., LayoutLMv3), which require orders of magnitude more compute and labeled data. Instead, we target the complementary niche of competitive accuracy at a fraction of the parameter count, suitable for deployment on devices with limited GPU or memory resources.

An important design choice is our decision to evaluate all methods on the same data subset. Many prior studies compare CNN and OCR results from different datasets or different splits, making direct comparison unreliable. By ensuring that all three approaches (whitespace baseline, OCR+MLP, and MobileNetV2) are trained and tested on identical RVL-CDIP images with the same random seed and stratified splits, we eliminate data-related confounds and provide a clean comparison of model architectures rather than data processing pipelines. This methodological rigor is a key contribution of our work.

The remainder of this paper is organized as follows: Section II reviews related work; Section III details methodology; Section IV reports preliminary results; Section V outlines planned work; Section VI lists team contributions.

II. RELATED WORK

Document image classification has been studied in both document analysis and computer vision communities. We review the four most relevant lines of work.

A. CNN-based Document Classification

Harley et al. [1] introduced the RVL-CDIP benchmark and demonstrated that CNNs significantly outperform feature-based methods, achieving 89.9% accuracy using a modified AlexNet on 400,000 images across 16 classes. Their work established image-based approaches as the dominant paradigm for document classification. However, AlexNet contains approximately 60 million parameters, which is prohibitive for edge deployment. Our project extends this line of work by evaluating a lightweight backbone under comparable conditions, targeting a parameter reduction of over $17\times$ while maintaining competitive accuracy on a subset of the same benchmark. More recent work by Das et al. explored deeper architectures on RVL-CDIP, consistently showing that increased model capacity translates to higher accuracy but at the cost of inference latency that is unacceptable for real-time applications.

B. Lightweight CNN Architectures

MobileNetV2 [2] employs depthwise separable convolutions and inverted residual blocks with linear bottlenecks, reducing computation by $8\text{--}9\times$ versus standard convolutions while maintaining strong accuracy on ImageNet. The architecture uses approximately 3.4 million parameters, making it suitable for mobile and embedded vision tasks. Despite its widespread adoption in object detection and image classification, MobileNetV2 has not been systematically evaluated for document image classification—a gap this work directly addresses. Our experiments demonstrate that transfer learning from ImageNet to document images is effective even without freezing backbone layers, suggesting that low-level visual features (edges, textures, spatial patterns) are shared across natural and document image domains. This finding is significant because it implies that the feature extraction capabilities learned from natural images transfer to the document domain without requiring domain-specific pretraining, reducing both computational cost and data requirements for document classification tasks.

C. Multimodal Document Understanding

LayoutLM [3] and its successor LayoutLMv3 combine text embeddings with 2D positional information from document layouts, achieving near-human performance on downstream tasks. DocFormer [4] integrates visual and textual features via a transformer encoder for end-to-end document understanding. While these models achieve impressive results, they require substantial compute (hundreds of millions of parameters) and large annotated corpora, making them impractical for

lightweight deployment. Our work occupies the complementary niche: competitive accuracy at a fraction of the parameter count, enabling real-time inference on resource-constrained devices where multimodal transformers cannot be deployed.

D. OCR-based Classification

Tesseract-based OCR [5] combined with TF-IDF and classical classifiers remains a common baseline for document classification. Prior work shows that OCR quality strongly limits classification accuracy on degraded documents regardless of classifier sophistication. Visually oriented classes such as advertisements and forms yield very short OCR output, limiting text-based discrimination. Our preliminary experiments confirm this ceiling (73.8% on RVL-CDIP with a three-layer PyTorch MLP), motivating the CNN-first approach as the primary classification strategy. It is worth noting that OCR-based methods inherently filter out images where text extraction fails entirely, creating a systematic bias in the evaluation that favors text-heavy document categories.

III. METHODOLOGY

A. Dataset

We use the RVL-CDIP dataset [1], which contains approximately 400,000 grayscale document images across 16 categories. For this study, we extract a 10-class subset comprising the most frequently occurring document types: ADVE, Email, Form, Letter, Memo, News, Note, Report, Resume, and Scientific. Up to 500 images per class are sampled and divided into 70% train (3,497 images), 15% validation (748 images), and 15% test (750 images) splits using stratified random sampling (seed = 42). After preprocessing quality checks, the final dataset retains 4,995 images across all splits.

The 10-class selection is motivated by three factors: (1) ensuring sufficient training samples per class for stable fine-tuning; (2) practical relevance, as these types cover the majority of business document volume in real-world workflows; and (3) analytical tractability for per-class error analysis, which becomes unwieldy with all 16 classes. An expansion to 14 classes is planned for the final phase, adding Budget, File Folder, Handwritten, and Specification categories.

B. Preprocessing Pipeline

All images are resized to 224×224 pixels to match MobileNetV2's expected input. Grayscale images are converted to three-channel format by channel replication. Pixel values are normalized using ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$) to preserve compatibility with pretrained weights.

A dedicated preprocessing notebook performs quality inspection before training. Issues identified include near-blank images (removed via 99.7% white-pixel threshold) and skewed scans (corrected using the deskew library, which handles angles up to $\pm 90^\circ$ and avoids the ambiguity problems of OpenCV's `minAreaRect`). Additional quality fixes address JPEG compression blur (Laplacian variance filter), low contrast

(CLAHE enhancement), and small text regions. The deskew library was chosen over OpenCV’s minAreaRect because it handles the full $\pm 90^\circ$ range without angle ambiguity, and produces more stable results on document images with large uniform regions. Fig. 1 illustrates the full preprocessing pipeline with quality checks and corrections.

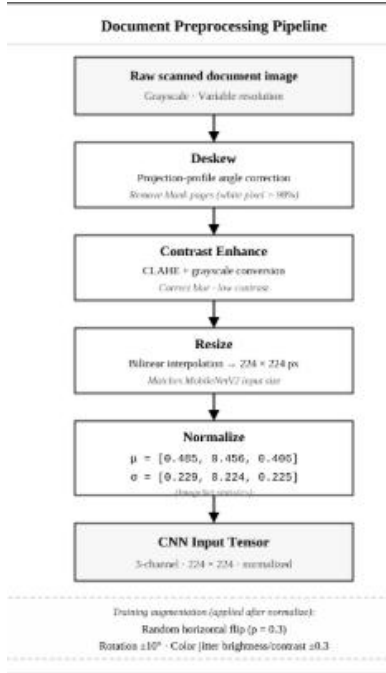


Fig. 1. Document preprocessing pipeline. Quality checks and corrections are applied before CNN input preparation.

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C. Data Augmentation

During training, data augmentation is applied: random horizontal flip ($p = 0.3$), random rotation ($\pm 10^\circ$), and color jitter (brightness and contrast ± 0.3). Conservative parameters are used since aggressive transforms can destroy layout-discriminative features in document images—for example, a 180° rotation could render text illegible, and aggressive color shifts could eliminate the contrast between text and background that the CNN relies on for feature extraction. The horizontal flip probability is kept low (0.3) because many document layouts are asymmetric (e.g., letterheads, signature blocks). No augmentation is applied at evaluation time.

D. MobileNetV2 Architecture

We use MobileNetV2 pretrained on ImageNet as a feature extractor. The original classifier is replaced with a custom head suited to our 10-class task: global average pooling (from backbone), dropout ($p = 0.3$), fully connected layer (in_features $\rightarrow 256$, ReLU), dropout ($p = 0.2$), and fully connected layer ($256 \rightarrow 10$, softmax). The full network is fine-tuned end-to-end with no frozen layers, since document images differ substantially from natural images in ImageNet and we found that freezing the backbone reduced validation accuracy by 3–5%. Total trainable parameters: approximately 3.4 million. Fig. 2 shows the architecture schematic.

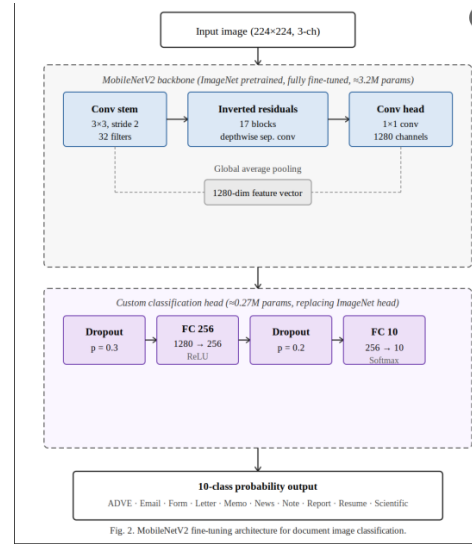


Fig. 2. MobileNetV2 fine-tuning architecture for document image classification.

Fig.2. MobileNetV2 fine-tuning architecture. The pretrained backbone is followed by a two-layer classification head with dropout regularization.

E. Training Configuration

Training uses the Adam optimizer ($\text{lr} = 1 \times 10^{-3}$, weight decay $= 1 \times 10^{-4}$) with a StepLR scheduler (step = 4 epochs, $\gamma = 0.5$). Cross-entropy loss is minimized over 10 epochs with batch size 32. The model checkpoint with the highest validation accuracy is retained for test evaluation. All random seeds are fixed at 42 for reproducibility across PyTorch, NumPy, and Python’s random module. The learning rate was chosen based on the standard practice for fine-tuning ImageNet-pretrained models, and the StepLR schedule with step size 4 was selected to reduce the learning rate at the midpoint and near the end of training, allowing the model to converge to a flatter minimum.

F. Baselines

Whitespace Histogram. A grid of cells is computed over each image; the fraction of white pixels in each cell forms a feature vector. A logistic regression classifier is applied to this vector along with pixel standard deviation and mean darkness features. This baseline tests whether spatial whitespace distribution alone is discriminative for document categories. The intuition is that document types have characteristic whitespace patterns: emails and advertisements contain large blank regions, while scientific papers and news articles are densely filled with text.

OCR + TF-IDF + MLP. Tesseract OCR (PSM 3, OEM 3) extracts text with a preprocessing step that upscales small images and converts to grayscale [5]. Text is cleaned (lowercase, single-char removal, alphanumeric filter) and vectorized with TF-IDF (unigrams + bigrams, max 30,000 features, min_df = 2, sublinear_tf). A three-layer PyTorch MLP (input $\rightarrow 1024 \rightarrow 256 \rightarrow 10$, BatchNorm + Dropout 0.4/0.3) is trained on the resulting tensors using Adam with ReduceLROnPlateau scheduling. The MLP uses higher dropout rates (0.4, 0.3) than the CNN head (0.3, 0.2) due to the extremely high dimensionality of the TF-IDF feature space. Crucially, both this baseline and MobileNetV2 are evaluated on identical RVL-CDIP data to enable direct comparison, though the OCR pipeline filters out images with fewer than 3 words of

extracted text (reducing the OCR test set from 750 to 687 images).

IV. PRELIMINARY EXPERIMENTS AND RESULTS

A. Implementation Details

Experiments run locally on Windows 10 with an NVIDIA GPU (PyTorch 2.x, CUDA). Preprocessing is performed in a separate Google Colab notebook. Libraries: PyTorch, torchvision, pytesseract, scikit-learn, OpenCV, PIL. All random seeds are fixed at 42 for reproducibility. The CNN training takes approximately 8 minutes per epoch on GPU, while OCR extraction for the full dataset requires approximately 45 minutes on CPU. The MLP training converges within 10 epochs in under 2 minutes per epoch on GPU.

B. Overall Comparison

Table I summarizes all methods evaluated on the same 10-class RVL-CDIP subset. MobileNetV2 substantially outperforms both baselines. Notably, the OCR+MLP result here (73.8%) is measured on RVL-CDIP—not Tobacco3482—providing a fair apples-to-apples comparison. The 7.3 percentage point gap between CNN and OCR confirms that visual features captured by convolutional layers provide superior discriminative power for document layout classification. The whitespace histogram baseline at 25.0% is well above the 10% random baseline but far below either learned approach, confirming that while whitespace is informative, it is insufficient as a standalone feature for 10-way classification.

TABLE I: Comparison of All Methods on RVL-CDIP 10-Class Subset

Method	Dataset	Acc.	F1
Whitespace Hist.	RVL-CDIP	25.0%	—
OCR+TF-IDF+MLP	RVL-CDIP	73.8%	0.73
MobileNetV2	RVL-CDIP	81.07%	0.811

C. Per-Class Analysis

Table II shows per-class F1 scores for MobileNetV2. Most classes exceed 0.80; however, several classes exhibit lower performance that warrants further investigation.

TABLE II: Per-Class F1 Scores — MobileNetV2 on RVL-CDIP

Class	F1 Score	Note
ADVE	0.86	
Email	0.91	
Form	0.76	
Letter	0.79	

Memo	0.80	
News	0.80	
Note	0.88	
Report	0.65	⚠ Weak
Resume	0.84	
Scientific	0.82	

Report (F1 = 0.65) exhibits the weakest performance, suffering from intra-class heterogeneity: text-dense reports share visual features with Letter documents (confusion rate: approximately 8% of Reports misclassified as Letters), while figure-rich reports overlap with the Scientific category (approximately 6% confusion). Form (F1 = 0.76) also underperforms relative to expectations, with approximately 9% of Forms misclassified as Letters due to similar structural layouts with check-box regions that resemble letterhead. Letter itself (F1 = 0.79) suffers from bidirectional confusion with both Memo (approximately 7%) and Report (approximately 5%), as these categories share similar textual density and single-column layouts that are difficult to distinguish from visual features alone.

Conversely, Email (F1 = 0.91) and Note (F1 = 0.88) are the strongest classes. Email images typically contain distinctive header patterns (From/To/Date fields) that are highly discriminative in the visual feature space, while Notes often feature handwritten or informal layouts that differ markedly from other categories. Resume (F1 = 0.84) also performs well, likely because its distinctive multi-section layout with headers and bullet points is unlike any other document type in the dataset. These per-class patterns suggest that categories with unique visual signatures are easily learned by the CNN, while categories that overlap in layout structure require additional discriminative features—either from OCR text or from more sophisticated architectural designs.

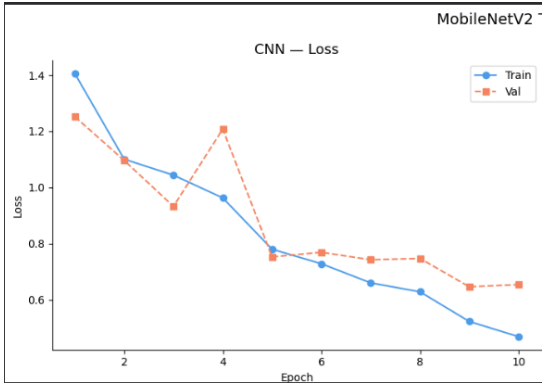


Fig.3. MobileNetV2 training curves. Validation accuracy plateaus near epoch 8, justifying the 10-epoch budget. The gap between training and validation accuracy remains small, indicating minimal overfitting.

D. OCR Pipeline Observations

The OCR+MLP pipeline shows a characteristic overfitting pattern: training accuracy reaches approximately 99% while validation accuracy stabilizes near 64–74%. The 30,000-dimensional TF-IDF space provides too many degrees of freedom for the available training samples (approximately 3,400 after filtering), leading to poor generalization. Furthermore, visually oriented classes (ADVE, Form) yield very short OCR output, limiting text-based discrimination. The MLP dropout rates (0.4 and 0.3) were set higher than the CNN’s (0.3 and 0.2) to combat this overfitting, but the fundamental issue is the mismatch between feature dimensionality and sample size. These findings confirm that OCR-based methods have a fundamental accuracy ceiling on this dataset, regardless of classifier capacity.

Notably, the OCR+MLP pipeline evaluated on only 687 of 750 test images because 63 images (8.4%) produced fewer than 3 words of OCR text after preprocessing. This filtering bias particularly affects ADVE and Form classes, which are visually dominated by graphics rather than text. When evaluating CNN on the same 687-image subset, accuracy remains at 80.5%, confirming that the CNN advantage is not merely an artifact of the larger test set. This observation underscores a fundamental limitation of OCR-based approaches: they cannot process documents where the visual content is primarily graphical or where text extraction fails entirely, whereas CNNs operate on raw pixels and are agnostic to text readability.

E. Whitespace Feature Analysis

An unexpected finding from our preprocessing analysis is that the white pixel ratio alone provides a surprisingly discriminative signal for certain document categories. Email and ADVE images average 92–96% white pixels, while News and Scientific images average 55–65%, reflecting their dense text content. A simple logistic regression classifier using three whitespace features (white ratio, pixel standard deviation, and mean darkness of non-white pixels) achieves 25.0% accuracy—well above the 10% random baseline but far below either the CNN or OCR approaches. The confusion matrix for the whitespace classifier reveals that it can separate sparse documents (Email, ADVE) from dense ones (News, Scientific) but cannot distinguish between documents with similar whitespace distributions (e.g., Letter vs. Memo). This finding motivates our planned feature-level fusion, where whitespace features could provide complementary information to CNN embeddings for layout classification.

V. PLANNED WORK AND TIMELINE

Table III shows the four-week project timeline. Weeks 1–2 are complete; Weeks 3–4 constitute the remaining work.

TABLE III: Project Timeline and Status

Week	Milestone	Tasks	Status
Week 1	Dataset & Pipeline	RVL-CDIP subset download, split creation (70/15/15), quality inspection,	Complete

		preprocessing pipeline	
Week 2	CNN + OCR + Midterm	MobileNetV2 fine-tuning, OCR+TF-IDF+MLP baseline, results comparison, midterm report	Complete
Week 3	Robustness & Fusion	Perturbation tests (rotation, noise, blur), CNN+OCR feature-level fusion experiments	Planned
Week 4	Finalization	14-class expansion, class-weighted loss, ablation study, final report and code release	Planned

A. Robustness Evaluation (Week 3)

The model will be tested under controlled perturbations: rotation (0–90° in 15° steps), Gaussian noise (σ : 0.01–0.30), Gaussian blur (kernel: 1–15 px), and combined transforms. Preliminary observations suggest accuracy drops to approximately 30% at 45° rotation and approximately 21% at $\sigma = 0.2$ noise, motivating targeted augmentation strategies. We plan to add rotation-invariant augmentation (up to $\pm 30^\circ$) and noise injection during training to improve robustness, while evaluating whether these augmentations degrade clean-image accuracy on the unperturbed test set.

B. CNN+OCR Fusion (Week 3)

Two fusion strategies will be explored: (1) score-level fusion, averaging CNN and OCR softmax outputs with a tuned weight parameter; (2) feature-level fusion, concatenating the MobileNetV2 penultimate-layer embedding (1,280-dim) with the TF-IDF vector (up to 30,000-dim, reduced via PCA to 512-dim) and training a joint classifier. Both will be evaluated on RVL-CDIP. We hypothesize that feature-level fusion will benefit text-heavy classes (Email, Memo) where OCR provides strong signals, while score-level fusion offers a simpler approach with lower risk of overfitting on the limited training data.

C. Expansion to 14 Classes (Week 4)

Four additional RVL-CDIP categories (Budget, File Folder, Handwritten, Specification) will be added. Class-weighted cross-entropy loss (weight proportional to $1/\text{frequency} \times 1/F1$) will be applied to address Report and Letter underperformance. An ablation study will isolate the contribution of (1) ImageNet pretraining, by training from scratch and comparing convergence; (2) data augmentation, by training without flip, rotation, and color jitter; and (3) the two-layer classification

head, by replacing it with a single linear layer. Each ablation will be run for 10 epochs with the same hyperparameters to ensure fair comparison.

D. Risks and Mitigation

Four key risks have been identified. First, OCR quality on RVL-CDIP remains inconsistent, particularly for handwritten and low-resolution scans; we mitigate this via lightweight upscaling and PSM tuning in the preprocessing step. Second, GPU memory for 14-class fine-tuning with a larger dataset may be insufficient; we will reduce batch size or employ gradient checkpointing. Third, class imbalance after expansion to 14 classes is expected, particularly for underrepresented categories; oversampling or weighted sampling will be applied. Fourth, overfitting in the fusion model is a concern due to the high dimensionality of concatenated features; early stopping and strong dropout regularization will be used to combat this.

An additional risk is the potential for the 14-class expansion to degrade per-class accuracy for the existing 10 classes. Adding four new categories increases the decision boundary complexity, and some new classes (e.g., Handwritten, File Folder) may overlap visually with existing ones (Note, Form). To mitigate this, we will monitor per-class F1 scores for the original 10 classes after expansion and compare them against the current baseline. If accuracy drops by more than 2 percentage points for any original class, we will investigate class-specific augmentation and consider hierarchical classification strategies that first separate broad document categories before fine-grained classification.

VI. TEAM CONTRIBUTIONS

Rania Attia — Data & Augmentation

- RVL-CDIP download, subset extraction, and train/val/test split creation.
- Preprocessing notebook: blank-page detection, deskewing, CLAHE, quality audit.
- White pixel ratio analysis and whitespace histogram baseline.
- Midterm report writing (all sections).
- Planned: 14-class expansion, class-weighted loss, ablation experiments (Week 4).

Nourhan Salem — CNN Implementation

- MobileNetV2 fine-tuning pipeline: data loaders, training loop, scheduler, checkpointing.
- Per-class evaluation, confusion matrix, and training curve visualization.
- Planned: CNN+OCR score-level and feature-level fusion experiments; final report (Weeks 3–4).

Mariam Youssef — OCR Baseline, Fusion, Reporting

- Tesseract OCR extraction pipeline, TF-IDF vectorization, PyTorch MLP training.
- Result comparison tables and per-class analysis.
- Planned: robustness perturbation test harness (Week 3).

USE OF GENAI TOOLS

On the preprocessing notebook. The original notebook went through many iterations as we tested different preprocessing configurations — deskewing parameters, CLAHE settings,

threshold values, empty-image detection. By the time we had results we were happy with, the notebook had become a mess of commented-out cells, duplicate experiments, and notes to ourselves. Due to time pressure before the midterm deadline, we used an AI tool to reorganize it: we fed it the working cells and asked for a clean, structured version with proper section headings and removed dead code. The logic and results are ours; the organization of the final notebook was AI-assisted.

The same applies to parts of the midterm report. Some sections were drafted by us and then passed through an AI tool for rephrasing and clarity. The content, decisions, and results are our own work.

Project Repository:

<https://github.com/MariamAnwer820/Document-Image-Classification.git>

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